Chapter 1

Quadrilaterals

motivation

1.1 Definition

A **quadrilateral is defined as the union of four unique noncollinear segments as follows:**

$$\square ABCD \coloneqq AB \cup BC \cup CD \cup DA.$$

Each segment is a side of the quadrilateral, and each point A, B, C, D is a vertex. Sides that share a vertex, such as AB and BC are adjacent, otherwise they are called opposite, such as AB and CD.

Theorem 1. *Each side of $\square ABCD$ contains its two endpoints and infinitely many interior points (by density).*

Theorem 2 (Adjacent sides intersect only at the common vertex)**.** $AB \cap BC = \{B\}$*, and similarly for the other three adjacent pairs.*

Proof. Suppose a point $X \neq B$ belonged to both AB and BC. By the definitions of segment and betweenness, this forces A, B, C to be collinear, contradicting the non-collinearity condition. $\square$

1.2 Diagonals

Definition 1. *Given a quadrilateral $\square ABCD$, the segments AC and BD are the* ***diagonals*** *of $\square ABCD$.*

1.3 Trapezoid/Trapezium

We define a trapezoid as a quadrilateral with at least one pair of parallel sides. (In some traditions, a trapezoid has exactly one pair; we choose the inclusive definition, which places parallelograms as a special case of trapezoids.)

Definition 2. *A convex quadrilateral $\square ABCD$ is a trapezoid if and only if*

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ or } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

1.3.1 Diagonals

1.3.2 Isosceles Trapezoid

A special case of the trapezoid occurs when the non-parallel sides (legs) are congruent. If this is the case, the quadrilateral is an isosceles trapezoid.

Definition 3 (Isosceles Trapezoid)**.** *A trapezoid* $\square ABCD$ *with* $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ *a isosceles trapezoid if and only if*

$$AD \cong BC.$$

Chapter 2

Quadrilaterals

motivation

2.1 Definition

A quadrilateral is defined as the union of four unique noncollinear segments as follows:

$$\square ABCD := AB \cup BC \cup CD \cup DA.$$

Each segment is a side of the quadrilateral, and each point A, B, C, D is a vertex. Sides that share a vertex, such as AB and BC are adjacent, otherwise they are called opposite, such as AB and CD.

Theorem 3. *Each side of $\square ABCD$ contains its two endpoints and infinitely many interior points (by density).*

Theorem 4 (Adjacent sides intersect only at the common vertex)**.** $AB \cap BC = \{B\}$*, and similarly for the other three adjacent pairs.*

Proof. Suppose a point $X \neq B$ belonged to both AB and BC. By the definitions of segment and betweenness, this forces A, B, C to be collinear, contradicting the non-collinearity condition. $\square$

2.2 Diagonals

Definition 4. *Given a quadrilateral $\square ABCD$, the segments AC and BD are the* ***diagonals*** *of $\square ABCD$.*

2.3 Trapezoid/Trapezium

We define a trapezoid as a quadrilateral with at least one pair of parallel sides. (In some traditions, a trapezoid has exactly one pair; we choose the inclusive definition, which places parallelograms as a special case of trapezoids.)

Definition 5. *A convex quadrilateral $\square ABCD$ is a trapezoid if and only if*

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ or } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

2.3.1 Diagonal

2.3.2 Isosceles Trapezoid

A special case of the trapezoid occurs when the non-parallel sides (legs) are congruent. If this is the case, the quadrilateral is an isosceles trapezoid.

Definition 6 (Isosceles Trapezoid)**.** *A trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ a isosceles trapezoid if and only if*

$$AD \cong BC.$$

2.3.3 Right Trapezoid

Definition 7 (Right trapezoid)**.** *A quadrilateral $ABCD$ is a* right trapezoid *if and only if it is a trapezoid and at least one of its interior angles is a right angle (if and only if at least one of its legs is perpendicular to the bases.)*

2.4 parallelograms

Theorem 5. *The diagonals of an isosceles trapezoid are congruent.*

Theorem 6 (Diagonals of isosceles trapezoid are congruent)**.** *Given and isosceles trapezoid $\square ABCD$ with $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$, the diagonals are congruent:*

$$AC \cong BD.$$

2.5 Special Parallelograms

2.5.1 Rectangles

2.5.2 Rhombus

Definition 8 (Rhombus is a parallelogram whose sides are all the same length)**.** *A rhombus is a parallelogram whose four sides are congruent to one another.*

Definition 9 (Rhombus is Equilateral but not Equiangular)**.** *me e9tuer Rhombus is Equilateral but not Equiangular*

2.5.3 Square

2.6 Parallelograms

Parallelogram A quadrilateral is a parallelogram if both pairs of opposite sides are parallel.

Definition 10 (Parallelogram)**.** *Let $A, B, C, D \in \mathbf{P}$ be four points satisfying the quadrilateral conditions. Then the quadrilateral $\square ABCD$ is a* ***parallelogram*** *if and only if*

$$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \text{ and } \overleftrightarrow{BC} \parallel \overleftrightarrow{DA}.$$

i hate this project lmao ts straight retarded. idk what to do with it ever.

2.7 Special Parallelograms

2.7.1 Rectangles

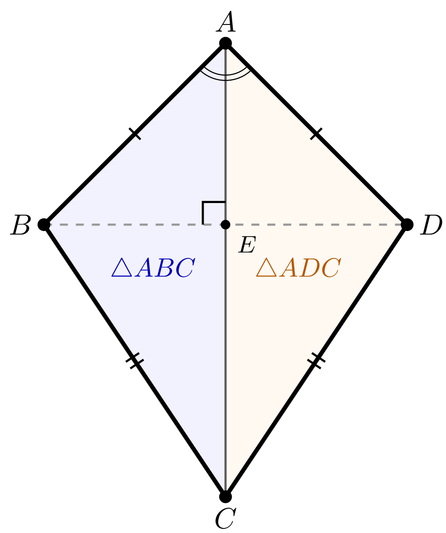
2.7.2 Rhombus

2.7.3 Square

2.8 Kites

Definition 11 (Kite)**.** *A convex quadrilateral $\square ABCD$ is a **kite** if*

$$AB \cong AD \wedge CB \cong CD.$$



Theorem 7. *In a kite $\square ABCD$,*

$$\angle BAC \cong \angle DAC \text{ and } \angle BCA \cong \angle DCA.$$

Proof. Consider the triangles $\triangle ABC$ and $\triangle ADC$.

- $AB \cong AD$ by definition
- $CB \cong CD$ by definition
- $AC \cong AC$ by reflexivity of congruence

By the SSS triangle congruency criteria, $\triangle ABC \cong \triangle ADC$. Consequently, the corresponding angles are congruent:

$$\angle BAC \cong \angle DAC \text{ and } \angle BCA \cong \angle DCA.$$

$\square$

Theorem 8 (The axis diagonal is the perpendicular bisector of the cross diagonal)**.** *In a kite $\square ABCD$, let $E = AC \cup BD$. Then,*

$$AC \perp BD \text{ and } BE \cong ED.$$

Proof. wasd □

Axiom 1 (Segment Extension with Betweenness). *For any A, B, C, D there exists E such that $H(A, B, C, E, D)$.*

Axiom 2 (Uniqueness of Extension). *If $H(A, B, C, E, D)$ and $H(A, B, C, E', D)$, then $E = E'$.*

Axiom 3 (Symmetry of Congruence). $H(A, B, C, D, E) \implies H(C, D, A, E, B)$

Theorem 9 (Midpoint quadrilateral of a kite). *The quadrilateral formed by joining the midpoints of the sides of a kite is a rectangle.*

Proof. □

2.8.1 Right Kite

2.9 Midpoint Theorems

2.10 Cyclic Quadrilaterals

2.11 Tangential Quadrilaterals

2.12 Bicentric Quadrilaterals

2.13 Other

11.15 Other Niche Quadrilaterals (Optional / Encyclopaedic) 11.15.1 Equidiagonal quadrilaterals (diagonals equal in length).

11.15.2 Orthodiagonal quadrilaterals (diagonals perpendicular).

11.15.3 Harmonic quadrilaterals (vertices form a harmonic division on the circumcircle – requires cross-ratio, so likely beyond your current primitives, but can be mentioned).

11.15.4 Quadrilaterals with special area properties (e.g., Brahmagupta's formula for cyclic quadrilaterals).

11.15.5 Complete quadrilaterals (the figure formed by four lines, six intersection points – a projective concept; may be out of scope).

11.15.6 Miquel point of a complete quadrilateral (four circles concurrent).